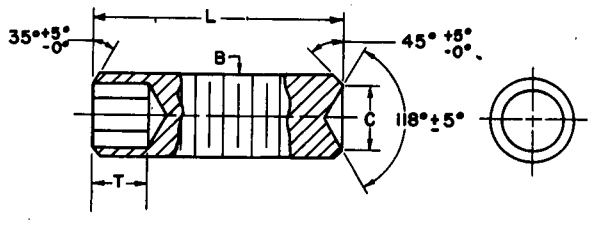
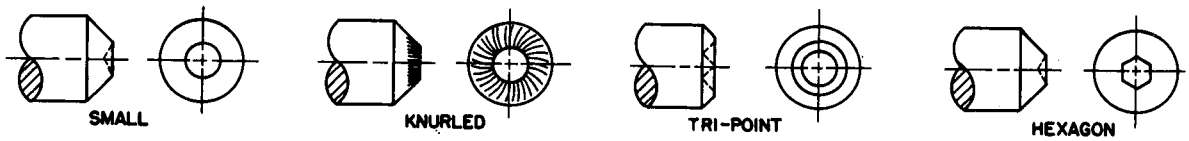
FED. SUP CLASS
5305





ACCEPTABLE CUP POINT DESIGNS

D NOMINAL SIZE		#0 (.060)	#2 (.086)	#4 (.112)	#6 (.138)	#8 (.164)
B THREADS PER INCH		80NF-3A	64NF-3A	48NF-3A	40NF-3A	36NF-3A
J WIDTH OF SOCKET ACROSS FLAT	Max	.0285	.0355	.051	.0635	.0791
	Min	.028	.035	.050	.0625	.0781
C DIAMETER OF POINT	Max	.033	.047	.061	.074	.087
	Min	.027	.039	.051	.064	.076
T KEY ENGAGEMENT	Min	.022	.028	.040	.050	.062

L LENGTH	Tolerance	Dash No.	FIIN	Dash No.	FIIN	Dash No.	FIIN	Dash No.	FIIN	Dash No.	FIIN
1/8 3/16 1/4 5/16 3/8 1/2	± 1-1/2% of L with a minimum of ± .010 in.	1* 2* 3*		9* 10* 11*		17 18 19 20 21		27 28 29 30 31		37 38 39 40 41 42	

D NOMINAL SIZE		#10 (.190)	1/4	5/16	3/8	1/2
B THREADS PER INCH		32NF-3A	28UNF-3A	24UNF-3A	24UNF-3A	20UNF-3A
J WIDTH OF SOCKET ACROSS FLAT	Max	.0947	.127	.1582	.1895	.2520
	Min	.0937	.125	.1562	.1875	.2500
C DIAMETER OF POINT	Max	.102	.132	.172	.212	.291
	Min	.088	.118	.156	.194	.270
T KEY ENGAGEMENT	Min	.075	.100	.125	.150	.200

L LENGTH	Tolerance	Dash No.	FIIN	Dash No.	FIIN	Dash No.	FIIN	Dash No.	FIIN	Dash No.	FIIN
3/16 1/4 5/16 3/8 7/16 1/2 5/8 3/4 1 1-1/4 1-1/2	± 1-1/2% of L with a minimum of ± .010 in.	48 49 50 51 52 53 54 55 56 57		63 64 65 66 67 68 69 70 71 72		78 79 80 81 82 83 84 85 86		92 93 94 95 96 97 98 99 100 101		107 108 109 110 111* 112*	

* Indicates manufacturer's non-stock production items.

NOTE: For material, protective coating and other pertinent data, see sheet 2.

Ⓐ CANCELLED
SEE SHEET 2

Ⓐ For changes see sheets 1 and 2

APPROVED 7 MAY 58 REVISED Ⓐ 29 JANUARY 1964

NOTICE.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A.	WC
Other Cust	SHIPS ASD

TITLE
SETSCREW, HEXAGON SOCKET, CUP POINT,
ALLOY STEEL, PHOSPHATE COATED, NF-3A AND UNF-3A

MILITARY STANDARD
MS51020

PROCUREMENT SPECIFICATION
FF-S-103

SUPERSEDES:

SHEET 1 OF 2

DD FORM 1 SEP 57 672-1 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

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KEY ENGAGEMENT: These dimensions apply to screws one diameter in length or longer. For screws shorter than one diameter in length, socket to be as deep as practicable.

MATERIAL: Steel, High-Grade Alloy, in accordance with Procurement Specification.

HARDNESS: Setscrews shall be hardened by heat treatment to a hardness of Rockwell C45 to 53.

PROTECTIVE COATING: Phosphate Coated, in accordance with MIL-C-16232, Type II.

THREADS: The threads shall be in accordance with Screw-Thread Standards for Federal Services, Handbook H-28.

DIMENSIONS: All dimensions are in inches unless otherwise specified.

PART NUMBER: The MS part number consists of the MS number, plus the dash number. Example: MS51020-1.

NOTES: (1) Referenced documents shall be of the issue in effect on date of invitations for bid.
(2) This standard takes precedence over documents referenced herein.

Ⓐ CANCELLED AFTER 29 January 1964 . Use MS51964.

INTERCHANGEABILITY RELATIONSHIP - MS51964 can replace the cancelled screws of the same size. The cancelled MS51020 cannot always replace MS51964 and should be used until existing stock is depleted. Then use only the superseding MS screws of the same size for design and replacement.

Ⓐ CANCELLED
SEE NOTE

This document has been promulgated by the Department of Defense as the Military Standard to limit the selection of the item, product, or design covered herein in engineering, design and procurement. This standard shall become effective not later than 90 days after the latest date of approval shown.

APPROVED 7 MAY 58 REVISED Ⓐ FOR CHANGES SEE SHEETS 1 AND 2.

This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. WC Other Cust SHIPS ASD	TITLE SETSCREW, HEXAGON SOCKET, CUP POINT, ALLOY STEEL, PHOSPHATE COATED, NF-3A AND UNF-3A	MILITARY STANDARD MS51020	
PROCUREMENT SPECIFICATION FF-S-103	SUPERSEDES:	SHEET 2 OF	

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